

Phishing Email Investigation Report

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Incident Date: 4 Dec 2024

Report Date: 31 May 2026

Classification Label: CONFIDENTIAL - INTERNAL USE ONLY

Organization: Globex Manufacturing Ltd

1. Executive Summary

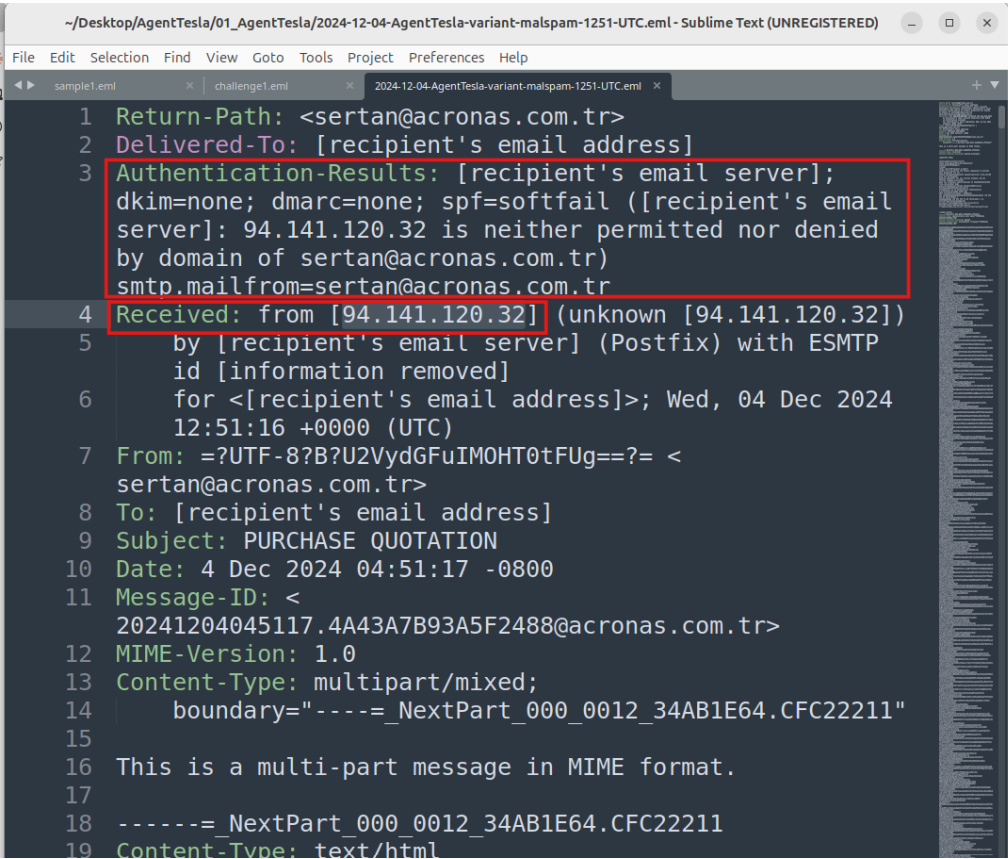
On December 4, 2024, a staff member reported opening an email attachment from a supplier. The investigation confirmed a highly targeted phishing campaign using a sophisticated, multi-layered evasion strategy. The threat actor masqueraded as a procurement professional from a real Turkish firm (acronas[.]com[.]tr) to deliver an advanced information-stealing payload. While network infrastructure and domain history appear "clean" on standard reputation scanners, deep analysis of the message headers, linguistic red flags, and attachment file structures conclusively proved a malicious intent to compromise corporate assets.

2. Email Analysis

2.1 Email Header Analysis

Field	Value
Sender Address	sertan@acronas[.]com[.]tr
Return-Path	sertan@acronas[.]com[.]tr

Originating IP	94.141.120.32
SPF Result	Softfail. The domain `acronas[.]com[.]tr` has an SPF record, but it does not explicitly authorize the originating IP `94[.]141[.]120[.]32` to send mail on its behalf.
DKIM Result	None
DMARC Result	None
Spoofing Detected	Yes. Highly probable spoofing/phishing attempt.



```
~/Desktop/AgentTesla/01_AgentTesla/2024-12-04-AgentTesla-variant-malspam-1251-UTC.eml - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
sample1.eml challenge1.eml 2024-12-04-AgentTesla-variant-malspam-1251-UTC.eml
1 Return-Path: <sertan@acronas.com.tr>
2 Delivered-To: [recipient's email address]
3 Authentication-Results: [recipient's email server];
  dkim=none; dmarc=none; spf=softfail ([recipient's email
  server]: 94.141.120.32 is neither permitted nor denied
  by domain of sertan@acronas.com.tr)
  smtp.mailfrom=sertan@acronas.com.tr
4 Received: from [94.141.120.32] (unknown [94.141.120.32])
5   by [recipient's email server] (Postfix) with ESMTP
   id [information removed]
6   for <[recipient's email address]>; Wed, 04 Dec 2024
   12:51:16 +0000 (UTC)
7 From: =?UTF-8?B?U2VydGFuIMOHT0tFUg==?= <
  sertan@acronas.com.tr>
8 To: [recipient's email address]
9 Subject: PURCHASE QUOTATION
10 Date: 4 Dec 2024 04:51:17 -0800
11 Message-ID: <
  20241204045117.4A43A7B93A5F2488@acronas.com.tr>
12 MIME-Version: 1.0
13 Content-Type: multipart/mixed;
14   boundary="-----_NextPart_000_0012_34AB1E64.CFC22211"
15
16 This is a multi-part message in MIME format.
17
18 -----_NextPart_000_0012_34AB1E64.CFC22211
19 Content-Type: text/html
```

Figure 1: Raw email header with annotated authentication results

A direct, single-hop connection. The email originated from the sender's host at IP address 94[.]141[.]120[.]32 and was delivered directly to the destination environment's mail server ([recipient's email server] via Postfix). No intermediate secure email

gateways or third-party smarthosts were recorded. The IP address 94[.]141[.]120[.]32 is assigned to an organization name DGTL TECH UK LLP and the raw email header data revealed that the sender's display name was encoded using Base64, which when decoded identified the display name as Sertan ÇOKER.

2.2 Email Content Analysis

- **Social Engineering Technique:** The social engineering technique used was based on financial curiosity and business urgency. The attacker asks for a "PURCHASE QUOTATION" for an upcoming project, enticing the recipient with a potential business deal. The email body references an "attached file" containing technical specifications.
- **Tactics Used (urgency/manipulation):** The attacker used pretexting tactics, attacker attached a "purchase quotation" for a new project called "TECHNICAL SECIFICATIONS.TAR", tricking the recipient into opening the malicious attachment. The attacker also crafted a highly believable story, pretending to be a procurement manager from a real Turkish company.
- **Linguistic Red Flags:** The email body displays critical grammatical errors and anomalous phrasing common in automated or non-native translation schemes. Notable indicators include:
 - "Dea Sir," (Typo in opening greeting)
 - "attached Equipment" (Irregular middle-sentence capitalization)
 - "We plan the purchase the equipment" (Broken syntax)
 - "We are Turkish company." (Omission of proper articles)

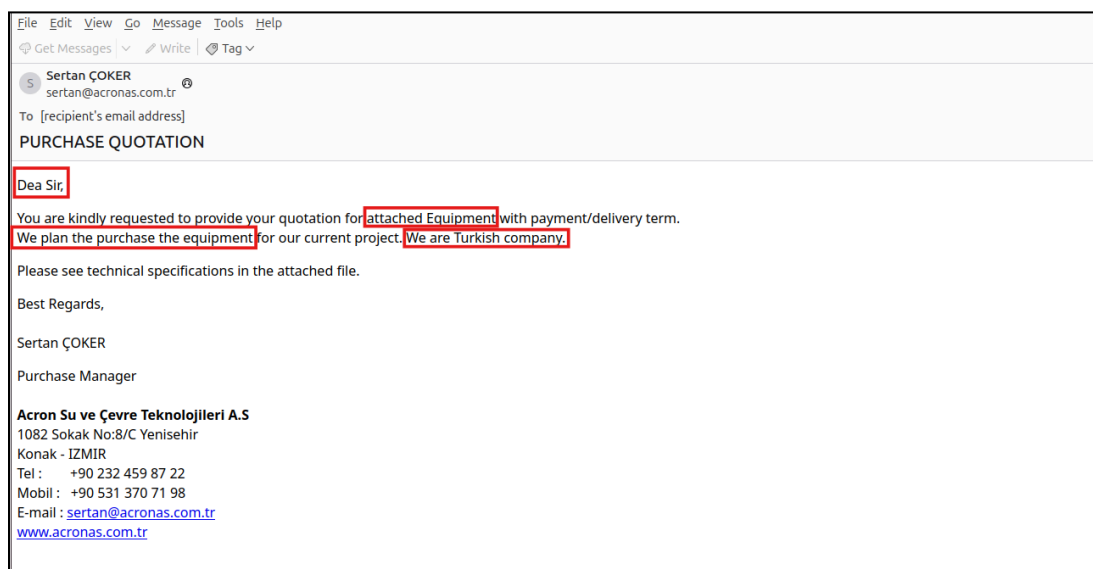


Figure 2: Email body as rendered in the victim's mail client, with annotated linguistic red flags indicative of a phishing attempt

3. URL Analysis

The URL `hxxp://www[.]acronas[.]com[.]tr` appears clean on VirusTotal and resolves to the serving IP address `94[.]73[.]145[.]234`. A reverse IP lookup revealed this is a shared server hosting at least three other domains: `akbulutiletisim[.]com`, `erfeyconsultancy[.]com`, and `sohbetroom[.]com`. The IP is physically located in Istanbul and owned by Cizgi Telekom A.S. When attempting to capture the site using URL2PNG, the request got stuck in an infinite loading loop, which can indicate the presence of anti-analysis scripts designed to hide the page content from automated scanners.

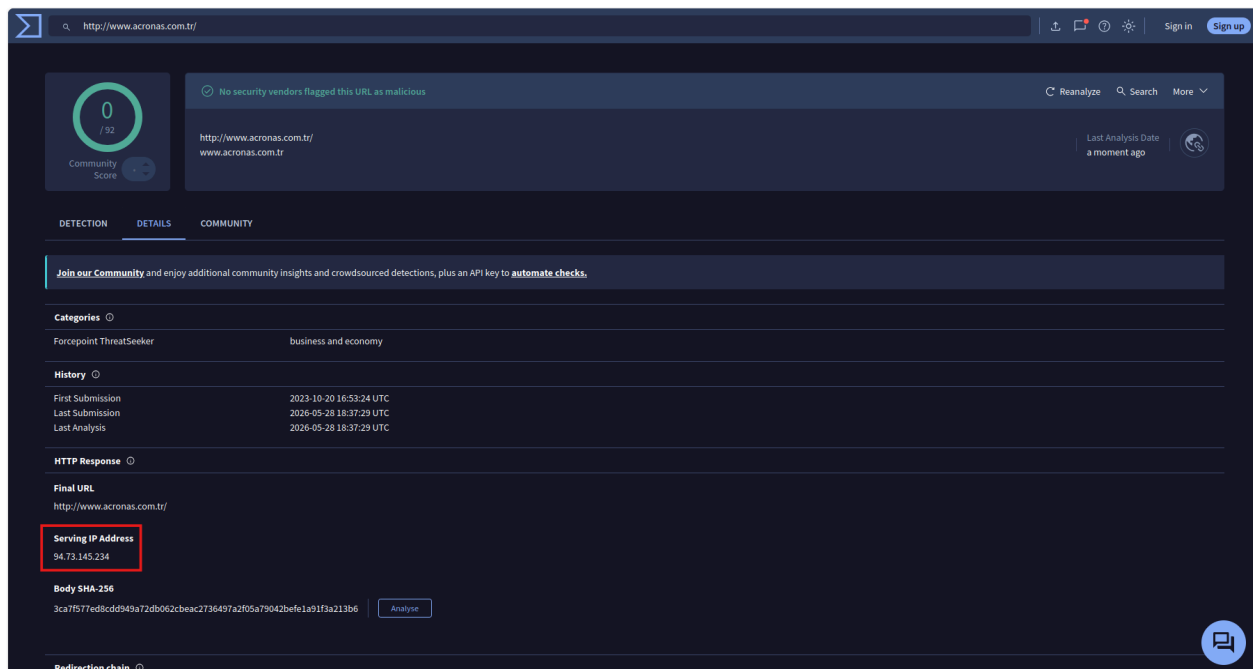


Figure 3: VirusTotal showing `hxxp://www[.]acronas[.]com[.]tr` serving IP address

DomainTools PROFILE CONNECT MONITOR SUPPORT Whois Lookup LOGIN Sign Up

Home > Reverse IP Lookup > 94.73.145.234

94.73.145.234 Reverse IP Lookup

Enter an IP address and our patented Reverse IP Lookup tool will show you all of the domains currently hosted there. Results include all gTLD domains and any known ccTLD domains.

ONE-CLICK MONITORING
Create an **IP Monitor** to monitor future changes to "94.73.145.234".
[Log in or Open an Account](#)

Lookup Connected Domains [Lookup tips](#)

94.73.145.234 **LOOKUP**

Example: 65.55.53.233 or 64.233.161.%

Reverse IP Lookup Results — more than 3 domains hosted on IP address 94.73.145.234

Domain	View Whois Record	Screenshots
1. akbulutletism.com	☐	
2. erleyconsultancy.com	☐	
3. sohbetroom.com	☐	

AND additional domains...
You must [Log In](#), [Sign Up](#) or [Buy this Report for \\$49.00](#) to access the results of your search.

[Sign Up](#)

Related Tools

Reverse NS Lookup
Discover all the domain names currently hosted on any given name server.

Name Server Monitor
Monitor the daily activity of any name server and receive notification of all new and/or deleted domains.

Hosting History
View historical IP addresses, name servers, and registrars for any given domain name.

IP Explorer
Explore the range of all IP addresses and discover how any particular IP block is being utilized.

IP Monitor
Passively monitor additions and changes to registered domain names associated with an IP Address.

Bulk Parsed Whois
Submit a list of domain names, and receive a .csv file with parsed Whois records for the domains.

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Figure 4: Reverse IP lookup

4. Attachment Analysis

- File Type & Name:** The email headers and text present the attachment as a .tar file. However, VirusTotal analysis and Magic byte data reveal the payload is actually a compressed Win32 RAR Archive (v4). The filename is inferred as "TECHNICAL SPECIFICATIONS" based on the body text lure.
- Cryptographic Hashes:**
 - MD5: b7635c9cc63619099419c68a2bf0d390
 - SHA1: 98683c9ee69dd591473629d231aacb1020db91b4
 - SHA256: 5c98308c69c84a57214442e2cad9cf8f0fcd9bab8e6050f9915ac336b6f1d59f0
- Intent & Purpose:** The attachment serves as a deceptive Trojan Dropper masquerading as an innocent "TECHNICAL SPECIFICATIONS" document. When opened by a user, it unrolls an embedded Windows GUI .Net assembly executable (.exe). This payload is a confirmed variant of the AgentTesla family.
- Malware Indicators:**
 - MALWARE: Highly malicious file globally blacklisted across prominent antivirus databases.

- STEALER: Programmed to aggressively hunt, harvest, and exfiltrate browser cookies, local email databases, and stored passwords.
- TROJAN: Cloaked inside a business-critical pretext to manipulate the user into initiating execution.
- EVADER: Uses extension masquerading (.tar vs .rar) and anti-sandbox tactics to bypass gateway detection rules.

```
ini@ini-Virtual-Machine: ~/Desktop/AgentTesla/01_AgentTesla$ python3 ../Tools/emldump.py 2024-12-04-AgentTesla-variant-malspam-1251-UTC.eml
1: M      multipart/mixed
2:      923 text/html
3:      798831 application/x-tar (TECHNICAL SPECIFICATIONS.TAR)
```

Figure 5: Terminal output showing the attachment file name & type

```
ini@ini-Virtual-Machine: ~/Desktop/AgentTesla/01_AgentTesla$ sha256sum TECHNICAL\ SPECIFICATIONS.TAR
5c98308c69c84a57214442e2cadcf8f0fcdab8e6050f9915ac336b6f1d59f0 TECHNICAL SPECIFICATIONS.TAR
ini@ini-Virtual-Machine: ~/Desktop/AgentTesla/01_AgentTesla$ sha1sum TECHNICAL\ SPECIFICATIONS.TAR
98683c9ee69dd591473629d231aacb1020db91b4 TECHNICAL SPECIFICATIONS.TAR
ini@ini-Virtual-Machine: ~/Desktop/AgentTesla/01_AgentTesla$ md5sum TECHNICAL\ SPECIFICATIONS.TAR
b7635c9cc63619099419c68a2bf0d390 TECHNICAL SPECIFICATIONS.TAR
ini@ini-Virtual-Machine: ~/Desktop/AgentTesla/01_AgentTesla$
```

Figure 6: Cryptographic Hashes

49/63 security vendors flagged this file as malicious

5c98308c69c84a57214442e2cadcf8f0fcdab8e6050f9915ac336b6f1d59f0

TECHNICAL SPECIFICATIONS.TAR

Size: 780.11 KB | Last Analysis Date: 29 days ago

Popular threat label: trojan.msi/agenttesla | Threat categories: trojan

Security vendors' analysis	Detection	Family labels
AhnLab-V3	Trojan.Win.MalwareX-gen.C5702736	trojan (spy) MSIL.Noon.gyf
ALYac	GenVariant.Application.Stictor.240530	Trojan/MSIL.SnakeKeylogger
Arcabit	Trojan.Generic.D479AA25	Win32.MalwareX-gen [Misc]
AVG	Win32.MalwareX-gen [Misc]	HEUR/AGEN.L385404
BitDefender	Trojan.GenericKD.75082277	Win1.Packed.MalwareX-10038470-0
CTR	Rar.trojan.msi	Malicious (score: 99)
DeepInstinct	MALICIOUS	Trojan.PackedNET.3.150
Emsisoft	Trojan.GenericKD.75082277 (B)	Trojan.GenericKD.75082277
ESET-NOD32	MSIL/Kryptik.AMTY.Trojan	MSIL/Formbook.FQQ0tr
GData	Trojan.GenericKD.75082277	Detected

Figure 7: VirusTotal analysis result

5. IOC Table

IOC Type	Indicator Value	Description
IP Address	94[.]141[.]120[.]32	Source IP of the originating mail sender server
IP Address	94[.]73[.]145[.]234	Serving IP address that the sender domain resolves to, hosted on a shared server in Istanbul
Sender email address	sertan@acronas[.]com[.]tr	Sender email address used to deliver the phishing lure, associated with the compromised domain
Domain	acronas[.]com[.]tr	Compromised sender domain
File name	TECHNICAL SPECIFICATIONS.TAR	Malicious attachment filename used as a business lure to deceive the recipient into opening the payload
File Extension	.tar	File extension spoofed to appear as a TAR archive; magic byte analysis confirms the true file type is a Win32 RAR Archive (v4), used to evade file-type-based gateway filtering
URL	hxxp[:]//[.]acronas[.]com[.]tr	Domain URL referenced in the email signature, resolves to a shared server with suspicious anti-analysis behavior
SHA1 Hash	98683c9ee69dd591473629d231aacb1020db91b4	Secondary cryptographic hash of the malicious attachment for cross-platform threat intelligence matching
MD5 Hash	b7635c9cc63619099419c68a2bf0d390	Cryptographic checksum of the malicious attachment used to uniquely identify and block the payload across endpoint security platforms

SHA256 Hash	5c98308c69c84a57214442e2cad9f8f0fdbab8e6050f9915ac336b6f1d59f0	Primary cryptographic identifier of the AgentTesla payload.
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```
ini@ini-Virtual-Machine:~/Desktop/AgentTesla/01_AgentTesla$ python3 ../Tools/eioc.py 2024-12-04-AgentTesla-variant-malspam-1251-UTC
.eml
Extracted IP Addresses:
=====
94[.]141[.]120[.]32

Extracted URLs:
=====
http://www[.]acronas[.]com[.]tr

Extracted Headers:
=====
Return-Path: <sertan@acronas.com.tr>
Authentication-Results: [recipient's email server]; dkim=none; dmarc=none; spf=softfail ([recipient's email server]: 94.141.120.32
is neither permitted nor denied by domain of sertan@acronas.com.tr) smtp.mailfrom=sertan@acronas.com.tr
From: =?UTF-8?B?U2VydGFuIM0HT0tFUG==?= <sertan@acronas.com.tr>
To: [recipient's email address]
Subject: PURCHASE QUOTATION
Date: 4 Dec 2024 04:51:17 -0800
Message-ID: <20241204045117.4A43A7B93A5F2488@acronas.com.tr>

Extracted Attachments:
=====
Filename: TECHNICAL SPECIFICATIONS.TAR
MD5: b7635c9cc63619099419c68a2bf0d390
SHA1: 98683c9ee69dd591473629d231aacb1020db91b4
SHA256: 5c98308c69c84a57214442e2cad9f8f0fdbab8e6050f9915ac336b6f1d59f0
```

Figure 8: Image showing the email IOCs

6. Risk Rating

- **Rating:** HIGH
- **Justification:** If an internal user successfully triggers the file execution, the threat actor will achieve silent persistence within the corporate host environment. The execution of an active InfoStealer exposes the enterprise to sweeping credential theft, unauthorized file access, and secondary lateral network penetration.

7. Initial Hypothesis of Attack

The attack was a calculated form of social engineering (pretexting) leveraging corporate procurement structures. By impersonating an environmental solutions manager from an established Turkish enterprise, the attacker intended to exploit commercial curiosity and financial opportunity. This psychological manipulation was designed to override a target's cybersecurity awareness, prompting them to open the

malicious archive container and execute the malware payload without their knowledge or consent.

8. Defense Actions

- Block sender email address and originating IP at the email gateway to prevent further delivery.
- Quarantine the malicious attachment hash across all endpoints to stop re-execution.
- Block FTP outbound traffic at the firewall.
- Enforce DMARC/DKIM/SPF policies at the mail gateway
- Alert all staff about this phishing campaign to reduce risk of other victims.
- Reset credentials of the affected user immediately to counter credential theft.
- Isolate the infected workstation from the network to contain the threat.
- Enable MFA on all user accounts to mitigate credential theft impact.
- Conduct regular phishing awareness training to reduce human risk